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→ "UPPER"
→ "LOWER"
PHOTONS

↑ ↓
SPIN-1/2 PARTICLE

2 SPIN ORIENTATION
↓
2 ALTERNATIVES



THE STATE OF A SPIN-1/2 PARTICLE CAN BE REPRESENTED BY A LINEAR COMBINATION OF TWO VECTORS OF A VECTOR SPACE

INDEED, WHEN MEASURING THE SPIN COMPONENT ALONG A CERTAIN DIRECTION U , ONE CAN GET ONLY TWO VALUES (EITHER $+\hbar/2$ OR $-\hbar/2$)

THUS ONE CAN WRITE ANY STATE OF THE PARTICLE AS $V = \alpha V_+ + \beta V_-$

$$P_U(+\hbar/2) = |\alpha|^2$$

$$P_U(-\hbar/2) = |\beta|^2$$

$$|\alpha|^2 + |\beta|^2 = 1$$

STATISTICALLY, IF I TAKE MANY COPIES OF THE PARTICLES AND I DO THE EXPERIMENT IDENTICALLY, I WILL GET A RESULT $+\hbar/2$ WITH A CERTAIN PROB. AND A RESULT $-\hbar/2$ WITH A CERTAIN PROB.

ONCE YOU CHOOSE A BASIS THEN YOU CAN ALSO REPRESENT ANY STATE USING TWO COMPLEX NUMBERS

E.g. $V = \alpha Z_+ + \beta Z_-$ REPRESENT THE STATE

$$V = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad Z_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$X_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$Y_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

THE VECTOR COMPONENT α AND β FOR A GIVEN STATE DEPENDS ON THE CHOSEN BASIS

E.g. LET US CONSIDER X_+

$$\{Z_+, Z_-\} \longrightarrow X_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\{X_+, X_-\} \longrightarrow X_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\{X_-, X_+\} \longrightarrow X_+ = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

THE ORDER OF THE ELEMENTS OF THE BASIS MATTERS

LET US NOW CONSIDER THE CASE IN WHICH THE PARTICLE IS DESCRIBED BY

$$V = \alpha Z_+ + \beta Z_-$$

WE WANT TO CALCULATE THE PROB. OF GETTING $-\hbar/2$ WHEN MEASURING S_y

$$V = \alpha Z_+ + \beta Z_- \quad S_y \quad P_Y(-\hbar/2) = ?$$

$$P_Y(-\hbar/2) = |\beta'|^2$$

$$V = \alpha' Y_+ + \beta' Y_-$$

$$\beta' = Y_-^\dagger V = \langle Y_- | V \rangle$$

$$\downarrow$$

$$Y_-^\dagger V = Y_-^\dagger (\alpha Z_+ + \beta Z_-) = \alpha Y_-^\dagger Z_+ + \beta Y_-^\dagger Z_- = \alpha \underbrace{Y_-^\dagger Z_+}_0 + \beta \underbrace{Y_-^\dagger Z_-}_1 = \beta'$$

$$\{Z_+, Z_-\} \quad V \rightarrow \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$Y_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$Y_-^\dagger V = \frac{1}{\sqrt{2}} (1, i) \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \frac{1}{\sqrt{2}} (\alpha + i\beta) = \frac{1}{\sqrt{2}} (\alpha + i\beta)$$

$$\beta' = \frac{1}{\sqrt{2}} (\alpha + i\beta)$$

REMEMBER: $w + w^* = 2\text{Re}(w)$

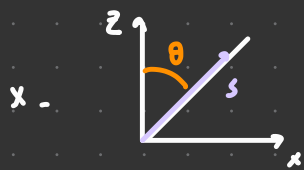
$$P_Y(-\hbar/2) = |\beta'|^2 = \frac{1}{2} (\alpha + i\beta)(\alpha^* - i\beta^*) = \frac{1}{2} [|\alpha|^2 + |\beta|^2 - \alpha\beta^* + i\alpha^*\beta] = \frac{1}{2} [|\alpha|^2 + |\beta|^2 + 2\text{Re}(i\alpha^*\beta)]$$

$$V = Z_+ \longrightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \alpha = 1 \quad \beta = 0 \quad P = 1/2 \quad \text{LET'S CHECK IT} \quad P_Y(-\hbar/2)_{Z_+} = \frac{1}{2} [1 + 0 + 2\text{Re}(0)] = 1/2$$

$$V = Y_- \longrightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \quad P = \frac{1}{2} \left[\frac{1}{2} + \frac{1}{2} + 2\text{Re}\left(i \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}}\right) \right] = 1/2$$

$$= \frac{1}{2} \left[\frac{1}{2} + \frac{1}{2} + 1 \right] = \frac{1}{2} \cdot 2 = 1$$

→ CONSIDER THE CASE IN WHICH THE PARTICLE IS IN THE STATE x_-



WHAT IS THE PROBABILITY OF GETTING $+\hbar/2$ WHEN MEASURING IN THE PLANE xz ALONG A DIRECTION INCLINED AT AN ANGLE θ FROM z -AXIS?

$$v = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$$

$$\{z_+, z_-\} \quad v = \cos \theta z_+ + \sin \theta z_-$$

$$\theta = 0 \rightarrow v = z_+$$

$$\theta = \pi/2 \rightarrow v = \cos(\pi/2) z_+ + \sin(\pi/2) z_- \\ = \frac{1}{\sqrt{2}} z_+ + \frac{1}{\sqrt{2}} z_- = x_+$$

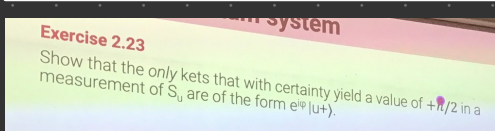
$$x_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$x_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

FIND $P_{\theta}(+\hbar/2) = ?$

$$= |v^\dagger x_+|^2 = \left| \begin{pmatrix} \cos \theta & \sin \theta \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right|^2 = \left| \frac{1}{\sqrt{2}} \cos \theta + \frac{1}{\sqrt{2}} \sin \theta \right|^2$$

$$\theta = \pi/2 \quad P_{\pi/2}(+\hbar/2) = \left(\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \right)^2 = 1$$



$$\{|u_+\rangle, |u_-\rangle\}$$

$$|v\rangle = \alpha |u_+\rangle + \beta |u_-\rangle$$

$$P_u(+\hbar/2) = 1 \leftarrow \text{WHICH GROUP OF KETS GIVE ME THAT?}$$

$$P_u(+\hbar/2) = |\langle u_+ | v \rangle|^2 = |\langle u_+ | (\alpha |u_+\rangle + \beta |u_-\rangle)|^2$$

$$= \left| \underbrace{\alpha \langle u_+ | u_+\rangle}_{\hookrightarrow 1} + \underbrace{\beta \langle u_+ | u_-\rangle}_{\hookrightarrow 0} \right|^2 = |\alpha|^2$$

ANY TIME I HAVE A COMPLEX NUMBER
WHOSE MODULUS SQUARED IS 1 I WILL
ALWAYS GET $+\hbar/2$
 $\hookrightarrow e^{i\theta}$

$$|v\rangle = e^{i\theta} |u_+\rangle$$

MEASUREMENTS AND FILTER

SUPPOSE WE HAVE A STERN-GERLACH APPARATUS ORIENTED ALONG THE z -AXIS



THE APPARATUS CAN BE USED TO MEASURE² THE SPIN OF THE INCOMING PARTICLES

² REMEMBER THAT HERE THE WORD MEASURE HAS A STATISTICAL MEANING

→ EXPECTED VALUE

NOW SUPPOSE THAT WE BLOCK ONE OF THE BEAMS (BLOCK THE ONE WITH SPIN $-\hbar/2$)



• IF WE PREPARE A PARTICLE IN THE STATE $|z_-\rangle$ AND SEND IT THROUGH OUR APPARATUS, THEN IT IS BLOCKED

• IF WE PREPARE IT IN $|z_+\rangle$ THEN IT WILL CERTAINLY PASS THROUGH THE APPARATUS

(HERE WE ASSUME THAT THE APPARATUS DOES NOT CHANGE THE STATE OF THE PARTICLE)

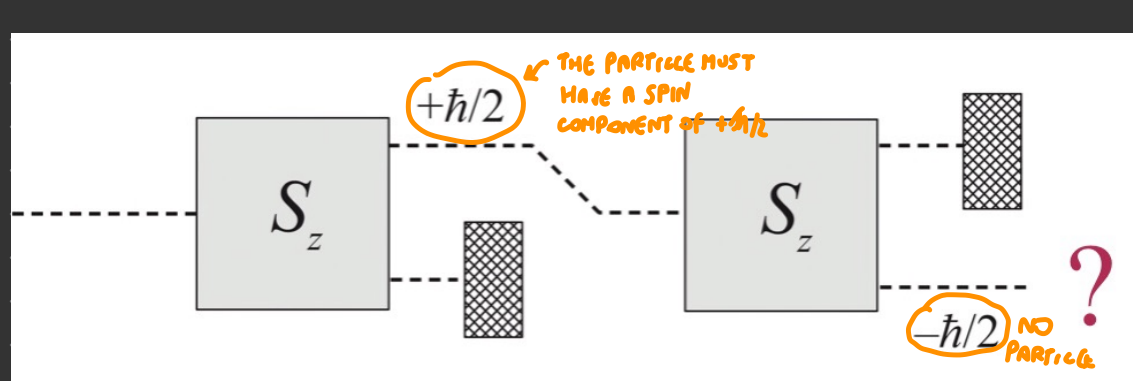
NOW SUPPOSE THAT WE START WITH THE STATE $|x_+\rangle$



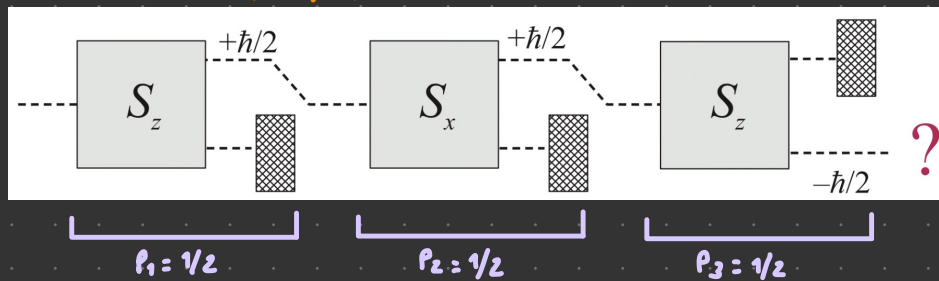
• WHAT IS THE PROBABILITY IT WILL PASS? $1/2$

• WHAT IS THE STATE AFTER S_z ? $|z_+\rangle$

NOW SUPPOSE THAT WE START WITH STATE $|x+\rangle$ AND CONSIDER THE SYSTEM



START WITH STATE $|x-\rangle$



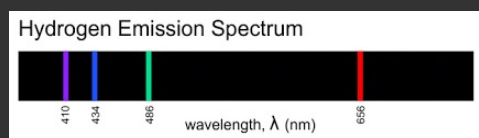
ARE WE GOING TO FIND ANY PARTICLE? MAY BE

• THE ORDER MATTERS

$$P_{\text{tot}} = P_1 + P_2 + P_3 = \left(\frac{1}{2}\right)^3 = 1/8$$

TWO LEVEL ATOMS

ATOM OF A GIVEN ELEMENT CAN ABSORB OR EMIT LIGHT ONLY AT CERTAIN DISCRETE FREQUENCIES (I.E. DISCRETE ENERGIES), THE PATTERN OF WHICH IS CHARACTERISTIC OF THAT ELEMENT



AN ATOM GENERALLY HAS MANY DIFFERENT ENERGY LEVEL.

IN MANY EXPERIMENTS, THOUGH, ONLY TWO ENERGY LEVELS - USUALLY THE GROUND STATE AND ONE EXCITED STATE - PLAY A SIGNIFICANT ROLE

E_1 ——— $|E_1\rangle$ EXCITED STATE

E_0 ——— $|E_0\rangle$ GROUND STATE

$$\langle E_0 | E_0 \rangle = \langle E_1 | E_1 \rangle = 1$$

IF WE SAY "THE ATOM IS THE GROUND STATE $|E_0\rangle$ " WE MEAN THAT IF WE MEASURE THE ENERGY OF THE ATOM, WE WILL GET THE VALUE E_0 W/ 100% PROBABILITY

THIS IS THE SAME SITUATION WE HAVE SEEN WITH THE SPIN $-1/2$ PARTICLE. IF WE SAY "THE PARTICLE IS IN THE STATE x " WE MEAN THAT IF WE MEASURE ITS SPIN COMPONENT ALONG x WE WILL ALWAYS GET $-1/2$

THE PRINCIPLE OF SUPERPOSITION TELLS US THAT THERE ARE OTHER STATES BE FORMED BY COMPLEX LINEAR COMBINATIONS LIKE

$$|\psi\rangle = \alpha_0 |E_0\rangle + \alpha_1 |E_1\rangle$$

$|\alpha_0|^2$ = PROB. OF FINDING THE ATOM IN STATE $|E_0\rangle$

$|\alpha_1|^2$ = PROB. OF FINDING THE ATOM IN STATE $|E_1\rangle$

EXTRA

TWO-LEVEL ATOMS

- THIS IS THE SAME SITUATION WE HAVE SEEN WITH THE SPIN-1/2 PARTICLE
- IF WE SAY: "THE PARTICLE IS IN THE STATE χ " WE MEAN THAT IF WE MEAS

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THE PRINCIPLE OF SUPERPOSITION TELLS US THAT THERE ARE OTHER STATES BE FORMED BY COMPLEX LINEAR COMBINATIONS LIKE

$$|\psi\rangle = \alpha_0|E_0\rangle + \alpha_1|E_1\rangle$$

$|\alpha_0|^2 = \text{PROB. of FINDING THE ATOM IN THE}$

